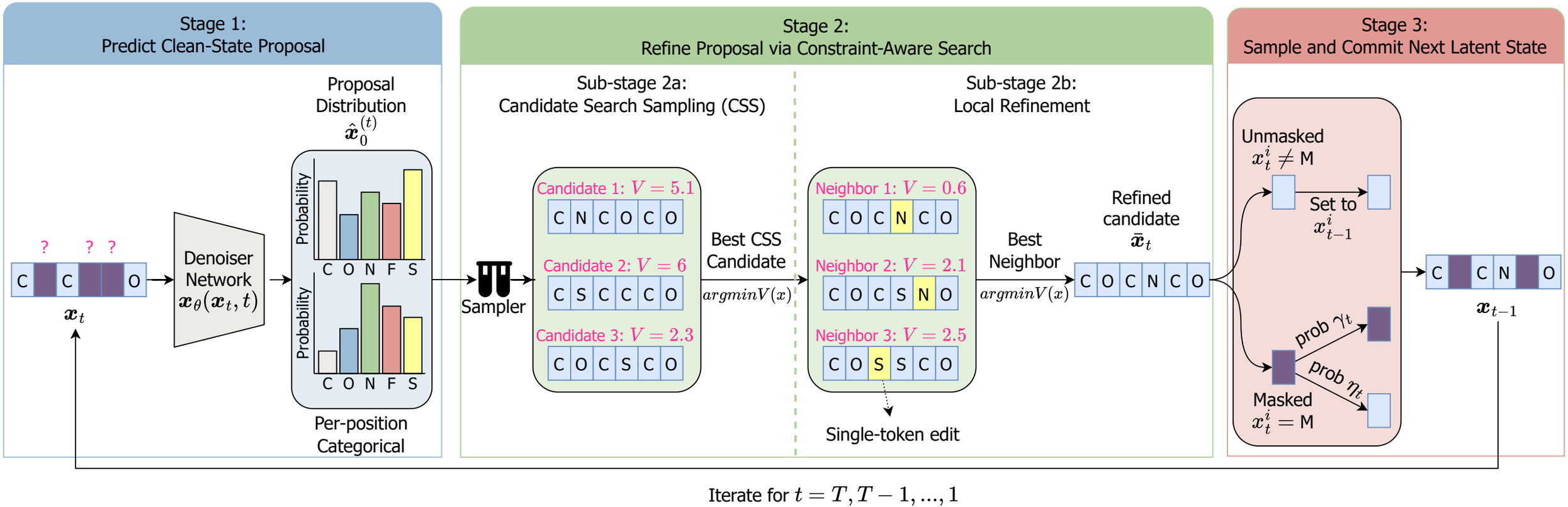
Stage 1:
Predict Clean-State Proposal
Proposal
Distribution
$\hat{x}_0^{(t)}$
Denoiser
Network
$x_\theta(x_t, t)$
x_t
Probability
C O N F S
Per-position
Categorical
Stage 2:
Refine Proposal via Constraint-Aware Search
Sub-stage 2a:
Candidate Search Sampling (CSS)
Sub-stage 2b:
Local Refinement
Sampler
Candidate 1: $V = 5.1$
Candidate 2: $V = 6$
Candidate 3: $V = 2.3$
Best CSS
Candidate
$argminV(x)$
Neighbor 1: $V = 0.6$
Neighbor 2: $V = 2.1$
Neighbor 3: $V = 2.5$
Single-token edit
Best
Neighbor
$argminV(x)$
Refined
candidate
$\bar{x}_t$
Stage 3:
Sample and Commit Next Latent State
Unmasked
$x_t^i \neq$ M
Set to
x_{t-1}^i
prob γ_t
prob η_t
Masked
$x_t^i =$ M
x_{t-1}
Iterate for $t = T, T-1, ..., 1$